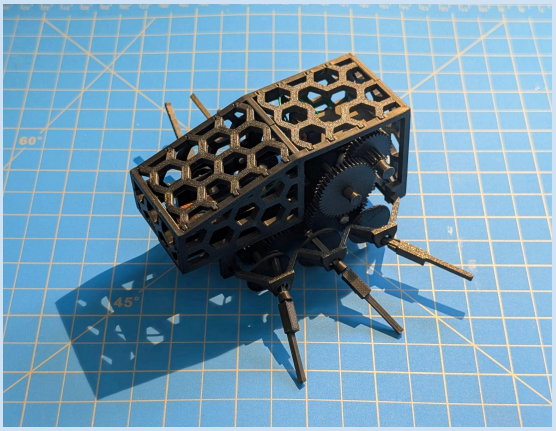
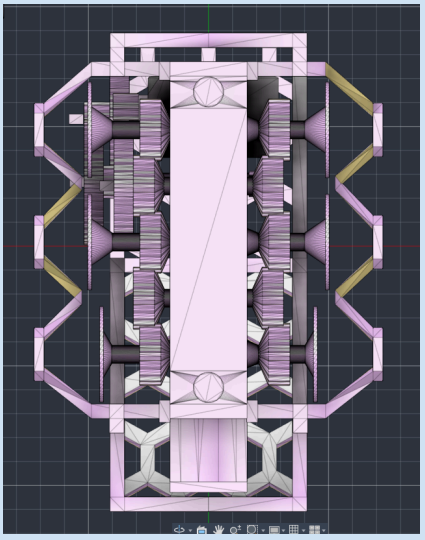
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Hexapod Rover

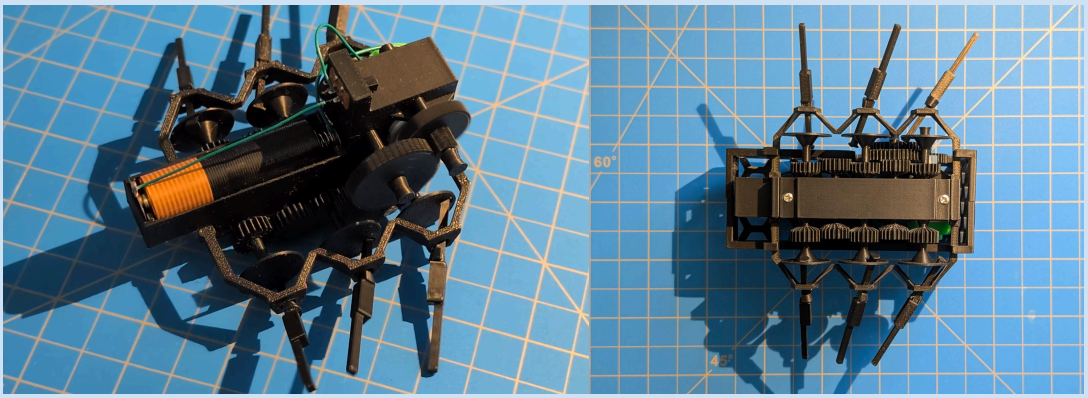


- **Fully conceptualized**, **designed**, and **constructed** a six-legged rover inspired by insect locomotion.
- Engineered a **fully custom spur gear drivetrain, exoskeleton, and linear actuators**.
- Achieved through FDM manufacturing.
- **Modular-centered design** process allows for quick repair and debugging.

Here is a short video demo.



- Designed in **Fusion360**, and simulated using **Blender** to visualize leg kinematics.
- The **gear system** required extensive testing and adjustments, to **find suitable tolerances**.
- The **custom gearbox** achieves a powerful **81:1** reduction ratio to **maximize torque**.
- Powered by a single **3V DC motor**.
- Achieving this amount of torque in a miniature design required careful **material and axle thickness adjustments**.

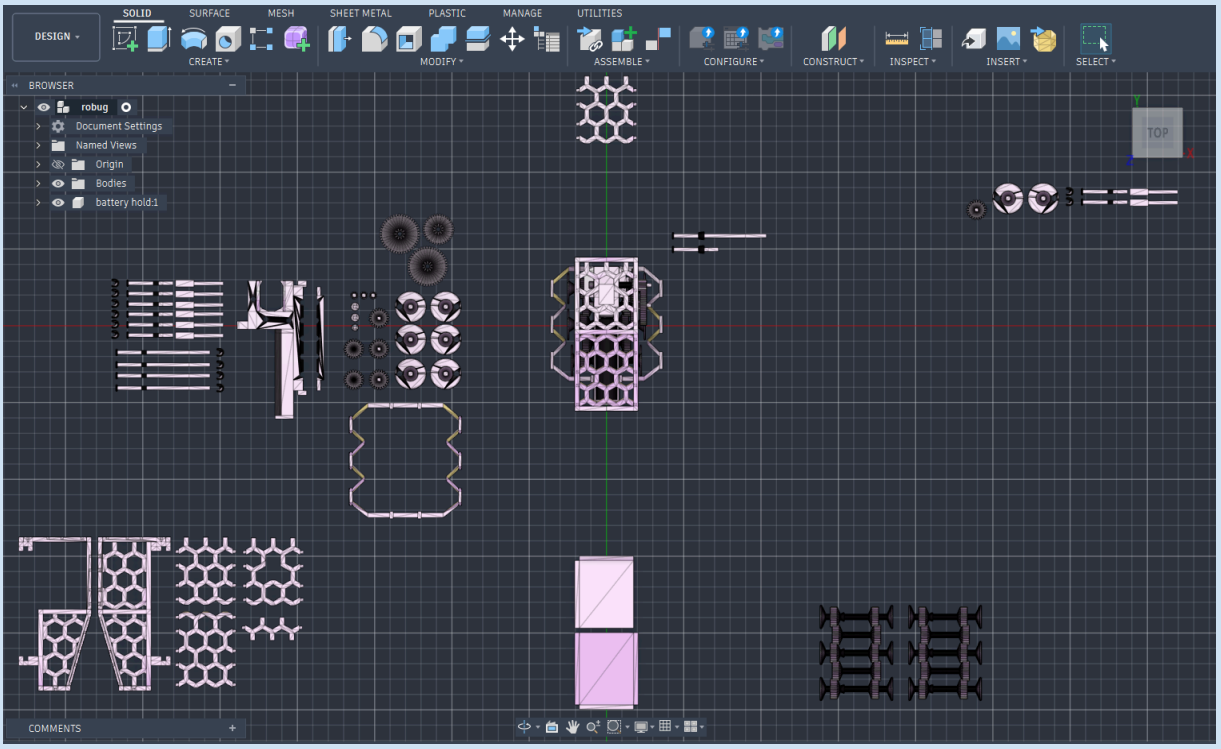


- The entire assembly is **modular**, making **troubleshooting more efficient**.
- This was especially necessary for the gear system and legs, which can be **easily disassembled** with screws.

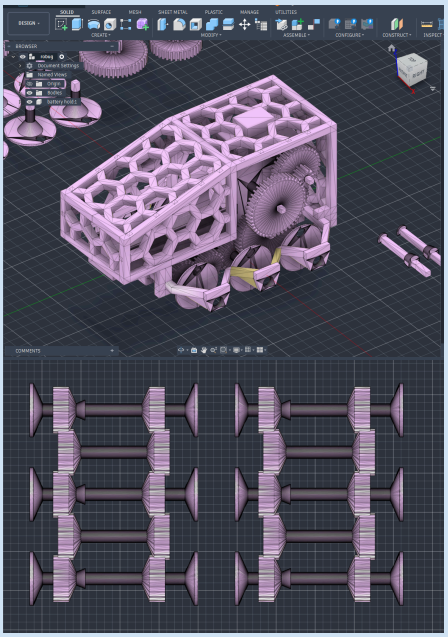
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[Here] is a **short video demo** of the **internal mechanisms** running in the rover.



A complete top-down view of the **CAD workspace**, with disassembled parts in **proper printing orientation** displayed.



The **exoskeleton** was **optimized to save material**, while acting as a **structurally sound shell**. It is also the pivot for the **linear actuator legs.**

The **greatest engineering challenge** during this project was perfecting the gear system. FDM manufacturing created relatively **huge limitations in tolerances and part quality**. This **severely limited gear fit and function**. The largest improvement arose from a unique solution: **horizontally-staggered gears.**

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The project went through **many design cycles** to troubleshoot errors. This gave rise to **many iterations** of the **Hexapod Rover.**

Right to left, top to bottom, images showcase regressing eras of the project's timeline.



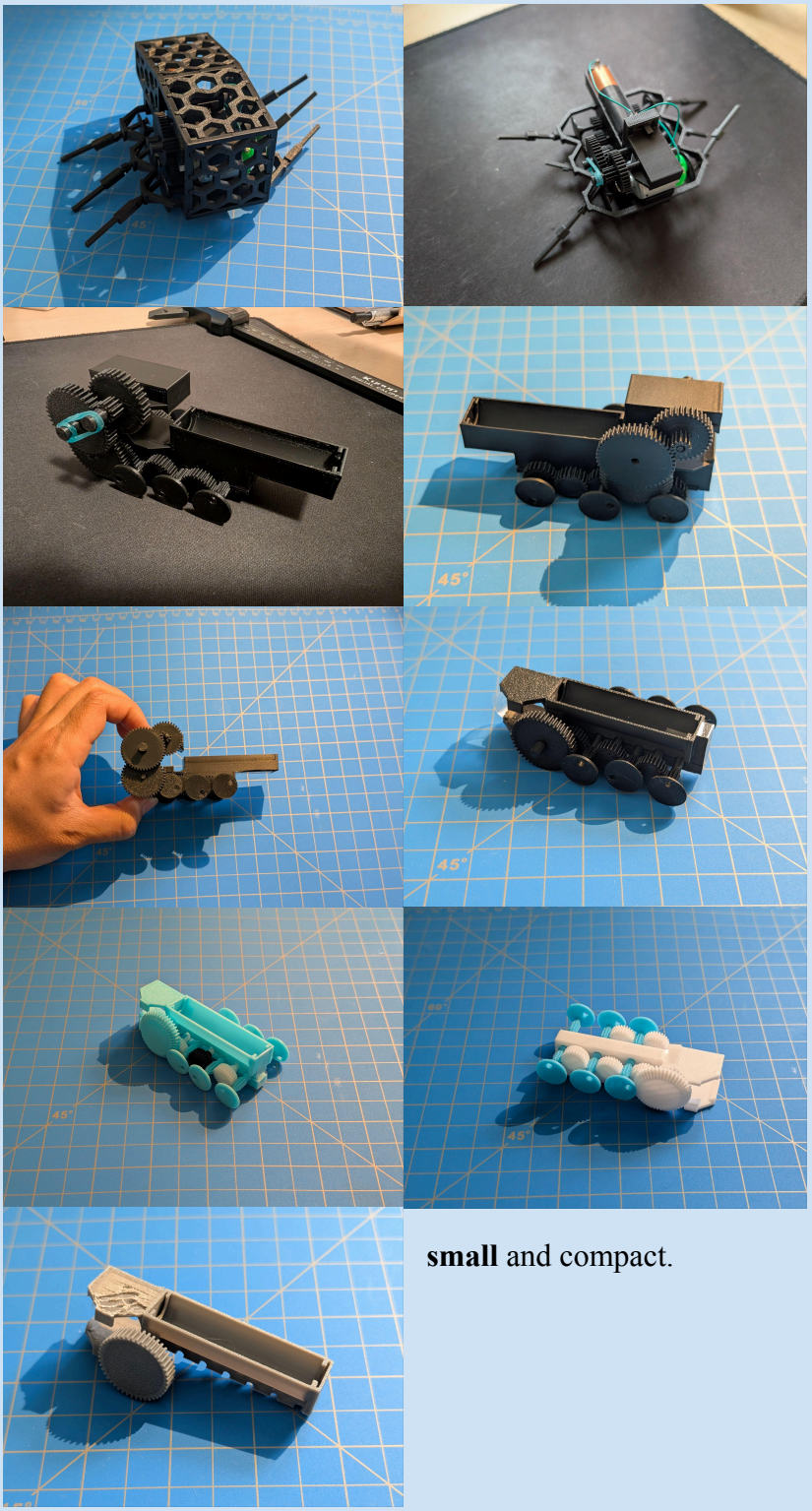
Changes were **constantly made** to fix issues with **tolerances**, **PLA strength**, **gear teeth**, and **gear fit**.

Isolated, larger challenges included designing the **linear actuators** for the legs, **motor placement**, and the creation of the **reduction gearbox**.

Moving the motor from the back to the front required **creative gear placements** to **keep the drivetrain**



The final model of the Hexapod Rover weighed in at **95 g**, with dimensions of **134 x 52 x 102 mm**.



small and compact.

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